

Charotar University of Science and Technology [CHARUSAT]
Faculty of Technology and Engineering
Department of Computer Science and Engineering
Course: CSE205 Data Communication & Networking
Practical Exam_Set2

Semester: 4th Sem B. Tech. (CSE)

Maximum Marks: 20

Date: 28/04/2025

Time: 9:00 a.m. to 12:30 p.m.

Evaluation Rubrics and Marks Distribution:

1. Practical Performance

Technical Execution (TE_x) [10 Marks]:

- Basic understanding: 1 mark
- Implementation Stage-1: 2–4 marks
- Implementation Stage-2: 5–7 marks
- Complete implementation Stage-3: 8–10 marks

Higher Order Thinking (HOT) [10Marks]:

- Basic problem-solving (Stage-1): 1- 5marks
- Advanced problem-solving and critical thinking(Stage-2): 6-10 marks

2. ViVa (Viva and Validation)

Technical Evaluation (TE_v) [4 Marks]:

- Partially accurate answers: 1-2 marks
- Accurate and conceptually sound answers: 3-4 marks

Communication Skills (CS) [1Marks]:

- Mixed language with moderate confidence: 0.5
- Fluent English communication: 1 marks

Note: While applying IP address, student need to allocate IP address as per his/her student ID. For Example, if student ID is 20DCS005 then IP address allocation for first network should start with 5.0.0.0. For subsequent network, it should start with ID+1 i.e. 6.0.0.0, 7.0.0.0. and so on.

Submission: After writing answer into this word document, Student need to change name to his ID followed by Set number. Ex 20DCS005_Set1.docx. Upload on assignment segment assigned on teams.

Part-A (10 marks)

you've been hired as a junior network engineer at BrightVision University, where the administration is upgrading its campus network. Your responsibility is to build and configure a reliable communication network between three key departments:

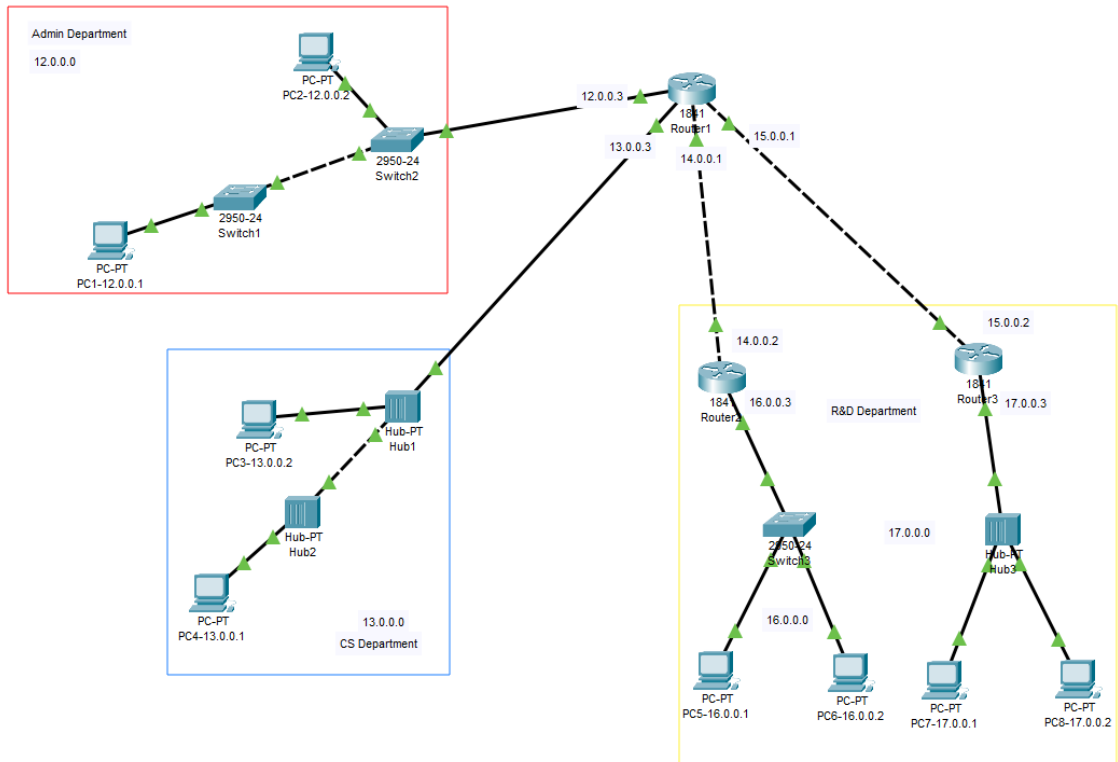
- Administration
- Computer Science
- Research & Development (R&D)

Each department is located in different buildings, and the university has provided basic networking devices and computers as shown in the layout below.

All department networks must be connected to a central router, which acts as the core of the university's communication system.

Your tasks include setting up the network, assigning IP addresses, configuring static routing so that every computer can talk to every other, and then verifying the network with ping and basic simulation analysis.

Device	Type	Connects To	Notes
PC1	PC	Switch1	Admin Department
PC2	PC	Switch2	Admin Department
PC3	PC	Hub1	Computer Science Department
PC4	PC	Hub2	Computer Science Department
PC5	PC	Switch3	R&D Department
PC6	PC	Switch3	R&D Department
PC7	PC	Hub3	R&D Department
PC8	PC	Hub3	R&D Department
Switch1	Switch	Switch2	Admin Network
Switch2	Switch	Router1	R&D Department
Switch3	Switch	Router 2	R&D Department
Hub1	Hub	Router1	CS Department (Legacy Equipment)
Hub2	Hub	Hub 1	R&D Department (Legacy Equipment)
Hub 3	Hub	Router 3	Admin Department Router
Router2	Router	Router 1	CS Department Router
Router3	Router	Router 1	R&D Department Router
Router1	Router	Router2, Router3	Central router for inter-department communication



Topology created as mentioned in the question. Here the 3 departments are isolated from each other. This increases abstraction and security between departments. The departments can communicate through routers.

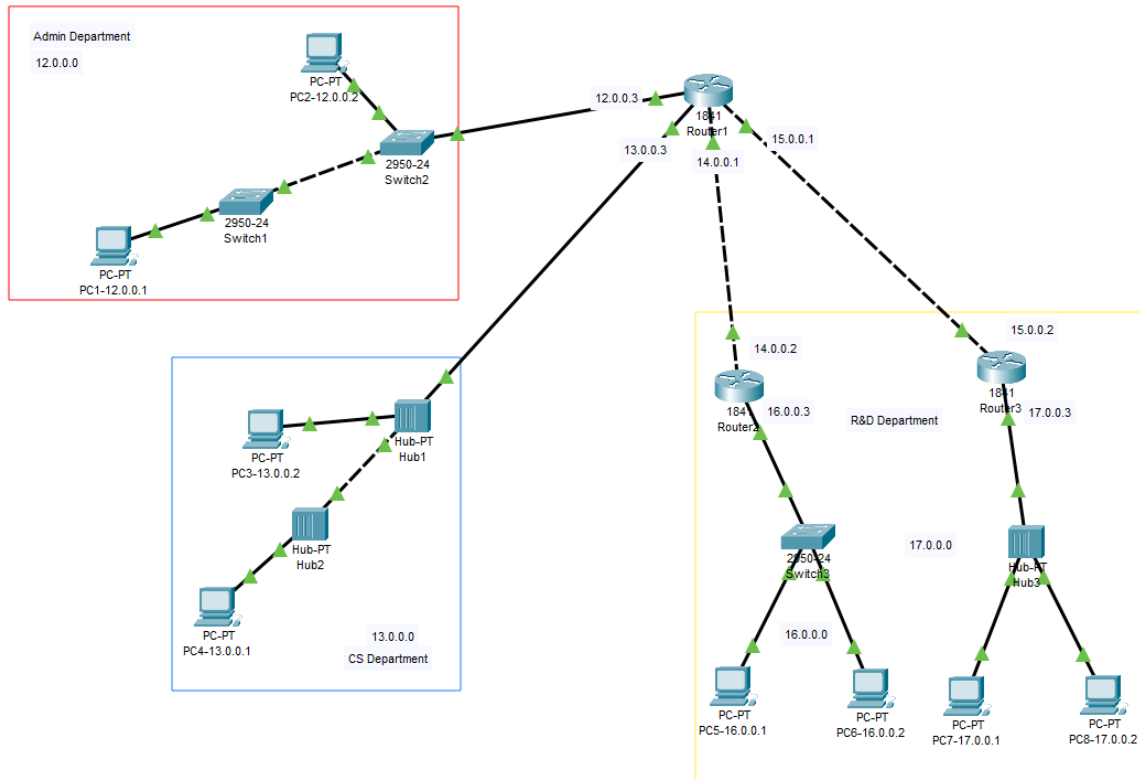
The Router 1 is the central router where all inter-department traffic goes through.

TASKS

Note: Students need to insert a image for each task and interpret the image in technical word.

Network Creation & Addressing (10 marks)

Stage 1:- Create the above topology in Cisco Packet Tracer.



- 1) Topology Is created according to the question.
The three departments are isolated from each other , i.e every department is in a separate buildings. Inter department communication can take place through static routing configured in the steps after.
- 2) For Admin and CS department , there is a tree-like topology for the end devices. And the R&D department adapts a star topology.
- 3) A combination of newer intermediate devices like switch that operated on the data-link layer , and older devices like hubs that operate only on physical layer are used.

Stage 2:- Assign IP addresses to PCs and router interfaces according to your ID-based rule.

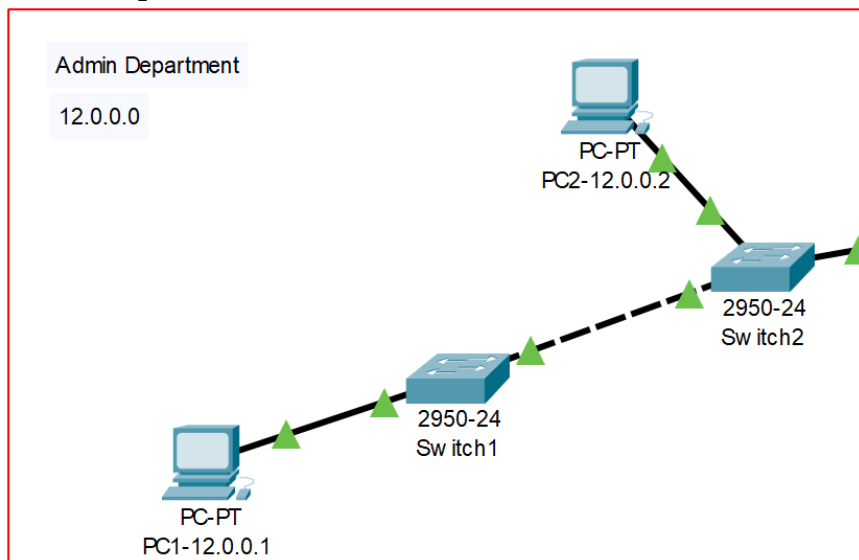
Here we can count all the subnets required and assign IP addressed based on the ID rule.

ID-23CS012

Admin Department	12.0.0.0
CS Department	13.0.0.0
Router 1 – Router 2 Link	14.0.0.0
Router 1 – Router 3 Link	15.0.0.0
R&D Department	16.0.0.0
R&D Department(Legacy)	17.0.0.0

Within Each Department:

Admin Department:

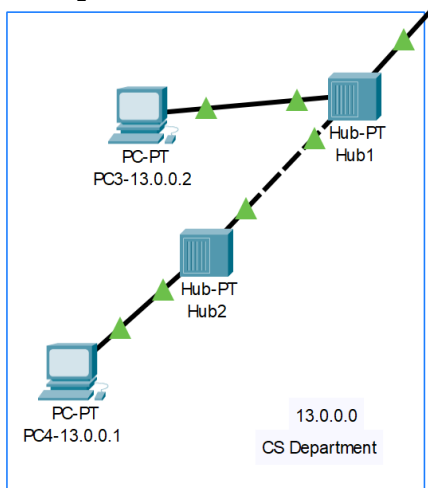


PC-1: 12.0.0.1

PC-2: 12.0.0.2

Gateway For All Devices(Router 1 Interface Address): 12.0.0.3

CS Department:

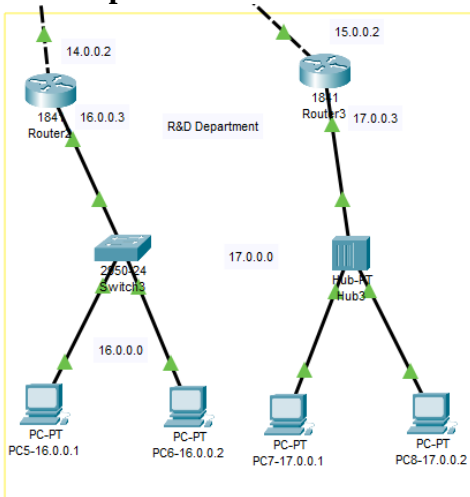


PC-4: 13.0.0.1

PC-3: 13.0.0.2

Gateway For All Devices(Router 1 Interface Address): 13.0.0.3

R&D Department:



PC-5: 16.0.0.1

PC-6: 16.0.0.2

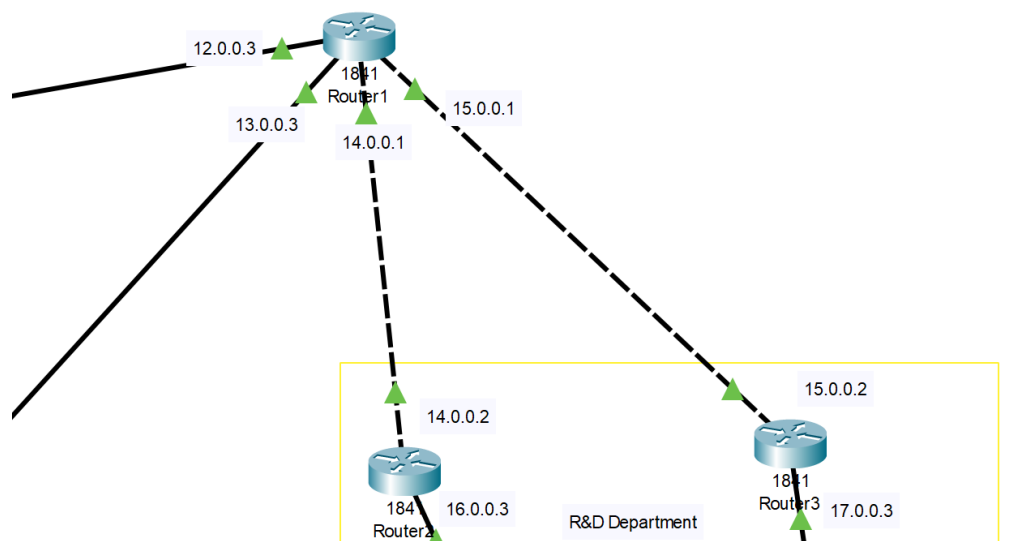
Gateway For PC5,PC6 (Router 2 Interface Address): 16.0.0.3

PC-7: 17.0.0.1

PC-8: 17.0.0.2

Gateway For PC7,PC8 (Router 3 Interface Address): 17.0.0.3

Router Links:



Router 1:

Interface – 1: 12.0.0.3 (Admin Dept)

Interface – 2: 13.0.0.3 (CS Dept)

Interface – 3: 14.0.0.1 (Router 2 Link)

Interface – 4: 15.0.0.1 (Router 3 Link)

Router 2:

Interface – 1: 14.0.0.2 (Router 1 Link)

Interface – 2: 16.0.0.3 (R&D Department)

Router 3:

Interface – 1: 15.0.0.2 (Router 1 Link)

Interface – 2: 17.0.0.3 (R&D Department (Legacy))

Interpretation:

IP Addresses are configured statically / manually for each End Device and router interface. it can be done by going in to config tab and setting the IPV4 address manually.

IP Addresses help to identify the which network (source) is sending data to where (destination) It is through ip addresses where routers configured to apply static routing. Routers create a routing table with IP addresses and next hop, to determine where the packet needs to be sent.

And as we can see all traffic going out of Router-2 and router-3 must go to router-1 which is the central router. Since It the only HOP available for data going outside the department.

Stage 3:- Implement static routing between routers to enable communication between all departments.

Implementing static routing:

Router 2 Configuration:

Network Address

12.0.0.0/8 via 14.0.0.1

13.0.0.0/8 via 14.0.0.1

15.0.0.0/8 via 14.0.0.1

17.0.0.0/8 via 14.0.0.1

As we can see all the traffic outside department is going to the central router 1 interface.

Router 3 Configuration:

Network Address

12.0.0.0/8 via 15.0.0.1

13.0.0.0/8 via 15.0.0.1

14.0.0.0/8 via 15.0.0.1

16.0.0.0/8 via 15.0.0.1

As we can see all the traffic outside department is going to the central router 1 interface.

Router 1 Configuration:

Network Address

16.0.0.0/8 via 14.0.0.2

17.0.0.0/8 via 15.0.0.2

As we can see router 1 directs 16.0.0.0 traffic to router 2 & 17.0.0.0 traffic to router 3

1. Fill the Tables After Configuration

(A) ARP Table for PC2 and PC8

ARP Table PC2:

Device	IP Address	MAC Address
PC1	12.0.0.1	0001.4318.b90b
Router 1	12.0.0.3	0000.0cec.bb02

ARP Table PC8:

Device	IP Address	MAC Address
PC7	17.0.0.1	00e0.f77b.4ec0
Router 3	17.0.0.3	00e0.f905.3101

(B) Switch Table for switch1 and switch3

Switch Name	MAC Address	Ethernet Port No
Switch1	0000.0cec.bb02	Fa0/1
Switch1	0001.4318.b90b	Fa0/2
Switch1	0050.0fb3.85c0	Fa0/1
Switch1	0060.5c2e.9601	Fa0/1
Switch3	0003.e449.ea02	Fa0/1
Switch3	00e0.a3e9.2ed2	Fa0/3
Switch3	00e0.f9a5.4daa	Fa0/2

(C) Router Static Routing Table

For each router

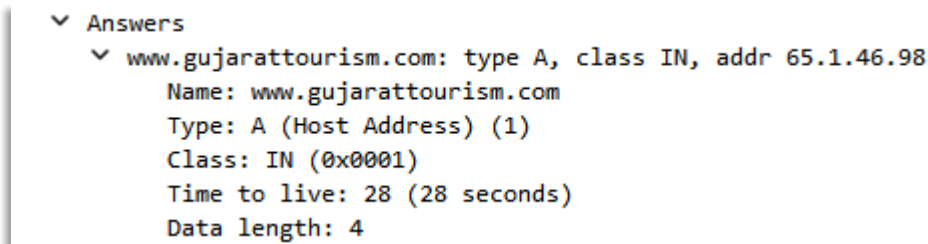
Router	Type	Destination Network ID	Next Hop IP Address	Hop Count
Router1	Static	12.0.0.0/8	directly connected	-
Router1	Static	13.0.0.0/8	directly connected	-
Router1	Static	14.0.0.0/8	directly connected	-
Router1	Static	15.0.0.0/8	directly connected	-
Router1	Static	16.0.0.0/8	14.0.0.2	2
Router1	Static	17.0.0.0/8	15.0.0.2	2
Router2	Static	12.0.0.0/8	14.0.0.1	2
Router2	Static	13.0.0.0/8	14.0.0.1	2
Router2	Static	14.0.0.0/8	directly connected	-
Router2	Static	15.0.0.0/8	14.0.0.1	2
Router2	Static	16.0.0.0/8	directly connected	-
Router2	Static	17.0.0.0/8	14.0.0.1	3
Router3	Static	12.0.0.0/8	15.0.0.1	2
Router3	Static	13.0.0.0/8	15.0.0.1	2
Router3	Static	14.0.0.0/8	15.0.0.1	2
Router3	Static	15.0.0.0/8	directly connected	-
Router3	Static	16.0.0.0/8	15.0.0.1	3
Router3	Static	17.0.0.0/8	directly connected	-

PART-B (5-marks)

Capture the traces of the given website in Wireshark & Perform the following tasks and also attach the image as required and interpret the images.

Website: <https://www.gujarattourism.com/>.

Stage 1:- (a) What is IP address of website?



Screenshot of wireshark analysis

Interpretation:

IP address can be obtained by analyzing the DNS response packets captured in Wireshark. We can filter the packets by filtering:

(dns.response_to)

packets which will show all the DNS response and we can search it there.

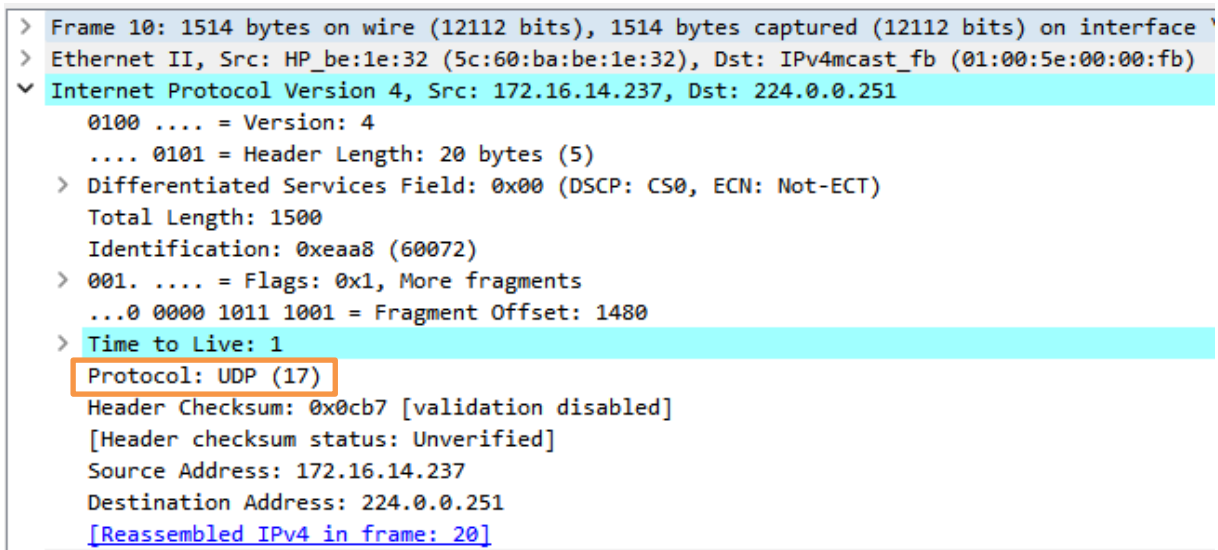
Better way would be to specifically search for the domain name in the raw packet data with frame matching filter:

frame matches “(www.gujarattourism.com)”

this will show only packets that contain the string “www.gujarattourism.com” in the payload.

THE IP ADDRESS IS: **65.1.46.98**

(b) Which transport layer protocol is utilized by packet number 10?



Interpretation::

Packet 10 can be analyzed by checking the no. column and selecting the packet.
From the packet analysis packet 10 is a fragmented IP packet with offset=1480, which was reassembled at packet 20.

The packet used the **UDP protocol** as seen by analyzing the packet IP header , and highlighted in image.

(c) Which version of Transport layer security used by website?

```
▼ Transport Layer Security
  ▼ TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
    Content Type: Application Data (23)
    Version: TLS 1.2 (0x0303)
    Length: 454
    Encrypted Application Data: 88a874db0a0b4a939c17a9b76de39bb1ec4721c4a473dd4b9bf519c61e5f4656a4b04b88...
    [Application Data Protocol: Hypertext Transfer Protocol]
```

ANSWER : TLS v1.2

Interpretation:

The website uses TLSv1.2 Protocol for security and data encryption. This is common practice for all **https** traffic.

TLS v1.2 encrypts the traffic from server to client and vice versa.

This is very important for security and it is the main reason for the adoption of https ,
Because standard http doesn't support TLS encryption which is why http websites are not considered secure in the current era.

Stage 2:- (a) What is Source MAC address and Destination MAC address for Packet no 10?

```
> Frame 10: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface
▼ Ethernet II, Src: HP_be:1e:32 (5c:60:ba:be:1e:32), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
  ▼ Destination: IPv4mcast_fb (01:00:5e:00:00:fb)
    Address: IPv4mcast_fb (01:00:5e:00:00:fb)
      .... ..0. .... = LG bit: Globally unique address (factory default)
      .... ..1. .... = IG bit: Group address (multicast/broadcast)
  ▼ Source: HP_be:1e:32 (5c:60:ba:be:1e:32)
    Address: HP_be:1e:32 (5c:60:ba:be:1e:32)
      .... ..0. .... = LG bit: Globally unique address (factory default)
      .... ..0. .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
```

Source: 01:00:5e:00:00:fb

Destination: 5c:60:ba:be:1e:32

Interpretation:

For packet no 10. The source and destination MAC addresses can be obtained by analyzing the Ethernet header which contains the source and destination MAC address fields.

(b) Calculate the time difference between packet no. 15 and packet no. 20 which was sent from your machine

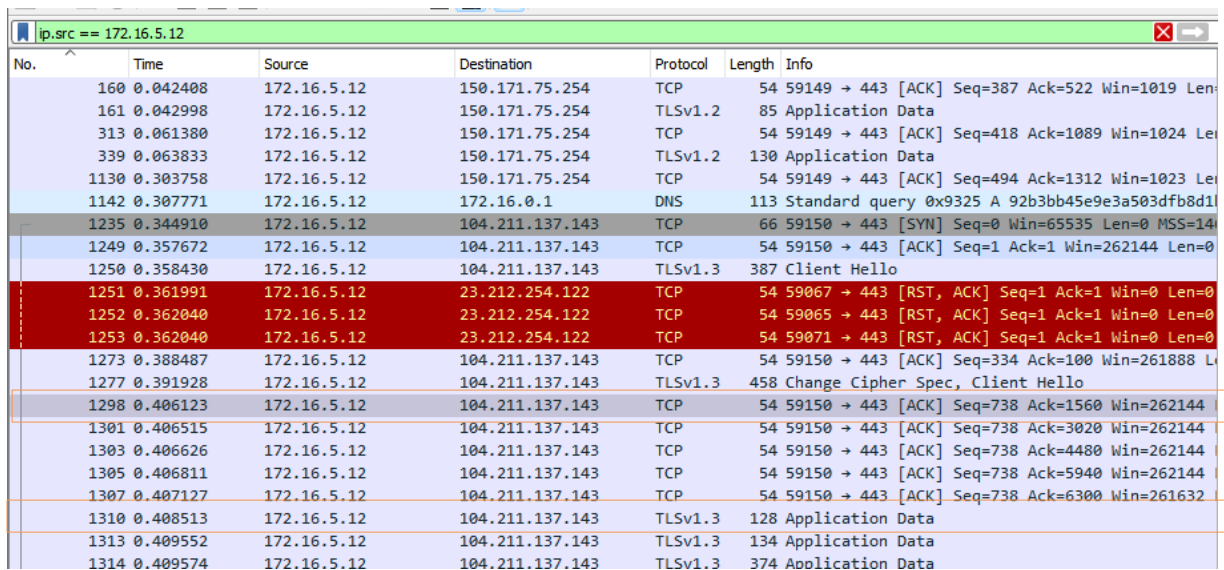
```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::161b:a15c:4676:395f%6  
IPv4 Address. . . . . : 172.16.5.12  
Subnet Mask . . . . . : 255.255.240.0  
Default Gateway . . . . . : 172.16.0.1
```

Finding IP address of my machine using ipconfig command.

Now we can filter the packets which have source ip address as 172.16.5.12 by using the filter

`ip.src == 172.16.5.12`

and then we can find the 10th and 15th packet sent by source ip = 172.16.5.12.



No.	Time	Source	Destination	Protocol	Length	Info
160	0.042408	172.16.5.12	150.171.75.254	TCP	54	59149 → 443 [ACK] Seq=387 Ack=522 Win=1019 Len=0
161	0.042998	172.16.5.12	150.171.75.254	TLSv1.2	85	Application Data
313	0.061380	172.16.5.12	150.171.75.254	TCP	54	59149 → 443 [ACK] Seq=418 Ack=1089 Win=1024 Len=0
339	0.063833	172.16.5.12	150.171.75.254	TLSv1.2	130	Application Data
1130	0.303758	172.16.5.12	150.171.75.254	TCP	54	59149 → 443 [ACK] Seq=494 Ack=1312 Win=1023 Len=0
1142	0.307771	172.16.5.12	172.16.0.1	DNS	113	Standard query 0x9325 A 92b3bb45e9e3a503dfb8d1
1235	0.344910	172.16.5.12	104.211.137.143	TCP	66	59150 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=14
1249	0.357672	172.16.5.12	104.211.137.143	TCP	54	59150 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
1250	0.358430	172.16.5.12	104.211.137.143	TLSv1.3	387	Client Hello
1251	0.361991	172.16.5.12	23.212.254.122	TCP	54	59067 → 443 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1252	0.362040	172.16.5.12	23.212.254.122	TCP	54	59065 → 443 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1253	0.362040	172.16.5.12	23.212.254.122	TCP	54	59071 → 443 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1273	0.388487	172.16.5.12	104.211.137.143	TCP	54	59150 → 443 [ACK] Seq=334 Ack=100 Win=261888 Len=0
1277	0.391928	172.16.5.12	104.211.137.143	TLSv1.3	458	Change Cipher Spec, Client Hello
1298	0.406123	172.16.5.12	104.211.137.143	TCP	54	59150 → 443 [ACK] Seq=738 Ack=1560 Win=262144 Len=0
1301	0.406515	172.16.5.12	104.211.137.143	TCP	54	59150 → 443 [ACK] Seq=738 Ack=3020 Win=262144 Len=0
1303	0.406626	172.16.5.12	104.211.137.143	TCP	54	59150 → 443 [ACK] Seq=738 Ack=4480 Win=262144 Len=0
1305	0.406811	172.16.5.12	104.211.137.143	TCP	54	59150 → 443 [ACK] Seq=738 Ack=5940 Win=262144 Len=0
1307	0.407127	172.16.5.12	104.211.137.143	TCP	54	59150 → 443 [ACK] Seq=738 Ack=6300 Win=261632 Len=0
1310	0.408513	172.16.5.12	104.211.137.143	TLSv1.3	128	Application Data
1313	0.409552	172.16.5.12	104.211.137.143	TLSv1.3	134	Application Data
1314	0.409574	172.16.5.12	104.211.137.143	TLSv1.3	374	Application Data

Then we can calculate the Time difference of both the packets

Time of packet 15 arrival = 0.406123s

Time of packet 20 arrival = 0.408513s

Time difference = 0.408513s - 0.406123s
= 0.00239s

